KOÇ UNIVERSITY
College of Engineering

COMP 438/538: INTRODUCTION TO REINFORCEMENT LEARNING

Midterm 1

Fall 2026, 05/11/2026
Duration: 100 Minutes

Name: Solutions

ID: 00110001 00110000 00110000

Read These Instructions Carefully

- This exam contains 14 pages including this cover page and 5 questions. Check to see if any pages are missing. Put your initials on the top of every page, in case the pages become separated.
- By submitting this exam, you **agree** to fully comply with Koç University Student Code of Conduct, and accept any punishment in case of failure to comply. If you do not submit this exam on time, you will receive 0 credits.
- The exam is **closed book** and **closed notes**. You are **not** allowed to use any additional material including any electronic equipment such as computers and mobile phones. An equation sheet is provided on the last page.
- You are expected to be able to provide clear and concise answers. Gibberish will not receive any credit. Your answers need to be readable by a human, illegible writing is not gradable.
- Show the essential calculations or a short justification. Exact fractions or equivalent decimals are acceptable. Each question describes a separate problem; state names are local to their question. Terminal states have zero continuation value, and a reward shown on a transition is the total reward for that transition.
- Read each question carefully and make sure to follow instructions. The written instructions take precedence over an answer that the instructor might give you during the exam, unless the instructor makes a class-wide announcement.
- Do not write in the table below.

Question:	1	2	3	4	5	Total
Points:	12	22	32	20	14	100
Score:						

1. (12 points) Short answers.

   (a) (2 points) Consider two policies for a continuing service task. One produces reward 0.75 every step. The other repeats the rewards $0, 2, 0, 2, \ldots$ forever. Which has the higher long-run average reward per step, and why? No average-reward learning algorithm is required.

   Solution: The alternating policy: $(0 + 2)/2 = 1$ per step versus 0.75 (1 for the rates, 1 for the comparison).

   (b) (4 points) A report compares MC and TD prediction on a fixed policy. It tries many step sizes and selects each method's best setting using the same episodes and seeds on which it reports final value error. It then uses the single lowest-error run to claim that the selected method is generally better. Identify two weaknesses and suggest a fix for each. No statistical test or numerical uncertainty calculation is required.

   Solution: (1) Settings were chosen on the runs used for the reported result, so the result is optimistic; repair: choose settings on separate development runs, then report fresh final runs at those settings. (2) One best run does not show typical performance; repair: run many independent runs and report the mean error with its variability, not the best seed. Rubric: 1 per weakness, 1 per repair. A fair-budget answer earns credit only if it is tied to one of these flaws.

For each statement below, write **True** or **False** and give a one-line justification. Each is worth 2 points: 1 for the verdict and 1 for the justification. A verdict without a justification receives 0.

(c) (2 points) Given q^*, a greedy optimal policy can be obtained without the model, however, obtaining one from v^* by one-step lookahead needs the model.

Solution: True: $\arg\max_a q^*(s,a)$ needs no model; with v^*, comparing actions needs the expected rewards and transition probabilities.

(d) (2 points) If the current policy is not optimal, one greedy policy-improvement step using its exact action values increases the value of at least one nonterminal state.

Solution: True: a suboptimal policy must admit a strict one-step gain somewhere; otherwise $v^\pi(s) = \max_a q^\pi(s,a)$ at every state, which is the Bellman optimality equation. By the policy improvement theorem the greedy policy keeps that gain and lowers no state. Rubric: verdict 1, reason for the strict gain 1, in ordinary language. Naming the policy improvement theorem without that reason earns 0.5 of the reason point; a change of action at a tie alone earns none.

(e) (2 points) The Monte Carlo prediction method we used in class can be applied directly to a continuing task that never terminates.

Solution: False: the method waits for a finished episode to observe its target, so without termination it never gets one. Rubric: verdict 1, reason 1. Full credit for explaining that a finite cutoff would change the target; "True, by truncating" earns no verdict point. Mentioning TD is not required.

2. (22 points) An online tutoring app runs a practice session of exactly 20 questions. Before each question it chooses a difficulty: easy, medium or hard. After each question it sees only whether the answer was correct and how many seconds it took. The student's actual skill is not visible, and it changes as the student practises. After the 20th question the student takes a fixed 10-question test that the app does not choose.

   The designer's objective is to maximize the student's expected score on the fixed final test. No model of how difficulty affects skill, correctness or response time is supplied.

   (a) (4 points) What are the actions? What does the app observe after each question?

   Solution: Actions: easy, medium, hard (2). Observation after each question: whether the answer was correct and the response time (2). Additional observed features (e.g. the topic) are fine if justified.

   (b) (6 points) Explain why the latest observation alone need not be a Markov state. Give two plausible histories ending in the same observation that could suggest different skill levels. Propose a history-based agent state for choosing the next difficulty, justify the information you retain, and mention one limitation. A justified approximation is sufficient; you do not need to prove that it is Markov.

   Solution: The skill is hidden and changes, so the latest answer alone does not summarize what matters for the next outcome (1). Example: "correct, 20 seconds" after ten correct hard answers versus after ten wrong easy answers suggests very different skill and a different next difficulty (2). Example agent state $S_t = f(H_t)$: recent counts of correct answers per difficulty (or an estimated skill level), plus questions remaining (2). Justification and one limitation, e.g. counts discard the order of answers, so improvement over the session can be missed (1).

(c) (6 points) Designer A rewards each correct practice answer with +1 and gives 0 otherwise. Designer B gives the final test score (0–10) after the session and 0 otherwise. Describe a plausible way A could encourage behaviour that does not improve the final test score. Which reward more directly matches the stated objective, and what makes learning from it difficult? You do not need to identify an optimal sequence of difficulties.

Solution: A can encourage choosing easy questions for immediate correct answers even when harder practice would teach more: an incentive mismatch, not a claim about the optimal policy for every student (2). B rewards the stated objective directly (2), but the feedback is delayed and sparse: one number after 20 decisions, so assigning credit to individual difficulty choices is hard and many sessions are needed (2).

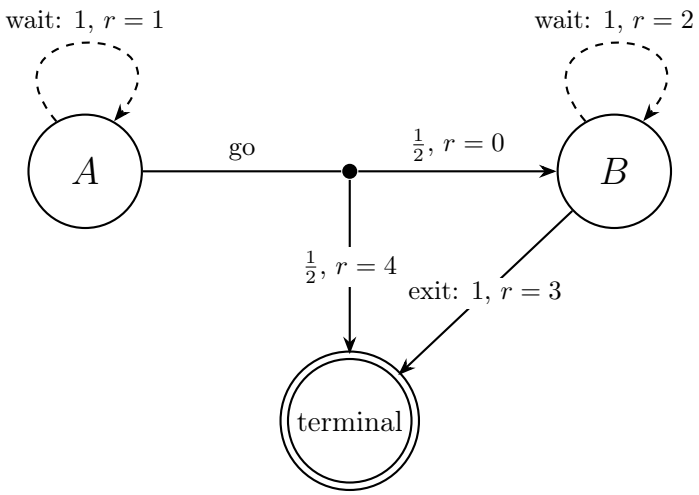
(d) (4 points) Is the task episodic or continuing? Should the number of questions remaining be part of the agent state? Why?

Solution: Episodic: the session ends after 20 questions and the test (1). Yes (1): with a deadline, the best difficulty can depend on the time left, e.g. hard practice may pay off only if enough practice follows, so the same skill estimate with 2 or 15 questions left can call for different actions (2). The "Yes" point does not require a separate counter: a state that already contains the remaining number of questions, e.g. the full history or the elapsed number of questions, earns it.

(e) (2 points) With no model of the student's responses or learning dynamics supplied, can you directly apply ordinary tabular value iteration to compute a policy? What information would be needed?

Solution: No (1): value iteration needs the expected rewards and next-state probabilities for a usable Markov state, which are not provided; experience, or an independently supplied model, would be needed (1). A conditional answer ("with a usable Markov model, yes") earns full credit.

3. (32 points) There are two nonterminal states A and B, and a terminal state. The discount is $\gamma = \frac{1}{2}$. The figure shows every possible transition. The black dot shows the possible outcomes after choosing go. Each arrow gives the probability $\mathcal{M}(s, a, r, s')$ and the reward r of its outcome. All other outcomes have probability zero.



The policy π chooses wait and go with probability $\frac{1}{2}$ each at A, and always chooses exit at B. Every sweep (i.e. an application of a step of an iterative algorithm) below is **synchronous**: all new entries use the previous sweep's table. Terminal continuation is always zero.

(a) (6 points) Starting from the expected-return meaning of v^π, write the two Bellman equations for this particular policy. Show where both the policy probability and the transition probability enter at A. Explain why the remaining return after reaching B can be replaced by $v^\pi(B)$. Do not solve the equations.

Solution:

$$v^\pi(A) = \tfrac{1}{2}\big[1 + \tfrac{1}{2}v^\pi(A)\big] + \tfrac{1}{2}\Big\{\tfrac{1}{2}\big[0 + \tfrac{1}{2}v^\pi(B)\big] + \tfrac{1}{2}\big[4 + \tfrac{1}{2}\cdot 0\big]\Big\}, \qquad v^\pi(B) = 3.$$

Equivalently $v^\pi(A) = \frac{3}{2} + \frac{1}{4}v^\pi(A) + \frac{1}{8}v^\pi(B)$. The outer $\frac{1}{2}$'s are $\pi(a \mid A)$; the inner $\frac{1}{2}$'s are the outcome probabilities of go. Given the next state B, the Markov property and continued use of π make the expected remaining return $v^\pi(B)$. Rubric: policy averaging 1, wait reward and continuation 1, averaging over go's outcomes 1, go's reward, discount and terminal handling 1, B equation 1, conditional-return explanation 1. (Exact solution, not required: $(5/2, 3)$.)

(b) (4 points) Starting with $V_0(A) = V_0(B) = 0$, perform two synchronous policy-evaluation sweeps for π. Fill in the table, showing each backup.

k	$V_k(A)$	$V_k(B)$
0	0	0
1	3/2	3
2	9/4	3

Solution: $V_1(A) = \frac{1}{2}(1+0) + \frac{1}{2}\left(\frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 4\right) = \frac{3}{2}$, $V_1(B) = 3$. $V_2(A) = \frac{3}{2} + \frac{1}{4}V_1(A) + \frac{1}{8}V_1(B) = \frac{3}{2} + \frac{3}{8} + \frac{3}{8} = \frac{9}{4}$, $V_2(B) = 3$. Rubric: 1 per cell (backup from the previous table 0.5, result 0.5). Carry a student's own earlier values forward; in-place updates keep the method credit but lose the affected result credit.

(c) (8 points) Start again from $V_0(A) = V_0(B) = 0$. Perform two synchronous **value-iteration** sweeps, showing both action-backup values at each state and identifying the maximum. Do not continue the table from part (b).

Sweep	A: wait	A: go	$V(A)$	B: wait	B: exit	$V(B)$
1	1	2	2	2	3	3
2	2	11/4	11/4	7/2	3	7/2

Solution: Sweep 2 from $(2, 3)$: A wait $1 + \frac{1}{2} \cdot 2 = 2$; A go $\frac{1}{2}\big(0 + \frac{1}{2} \cdot 3\big) + \frac{1}{2} \cdot 4 = \frac{11}{4}$; B wait $2 + \frac{1}{2} \cdot 3 = \frac{7}{2}$; B exit 3. The maximizing action at B changes from exit to wait. Rubric: for each state and sweep, both action backups 1 (0.5 each) and the maximum 1; four rows, 8 in total.

(d) (8 points) Independently of parts (b) and (c), the exact values of the original policy are supplied: $v^\pi(A) = \frac{5}{2}$ and $v^\pi(B) = 3$. Compute q^π for each of the four state–action pairs. State the greedy improved policy and check that its selected q^π value is at least v^π at each state.

Solution: $q^\pi(A, \text{wait}) = 1 + \frac{1}{2} \cdot \frac{5}{2} = \frac{9}{4}$; $q^\pi(A, \text{go}) = \frac{1}{2}\big(0 + \frac{1}{2} \cdot 3\big) + \frac{1}{2} \cdot 4 = \frac{11}{4}$; $q^\pi(B, \text{wait}) = 2 + \frac{1}{2} \cdot 3 = \frac{7}{2}$; $q^\pi(B, \text{exit}) = 3$ (1 each). Greedy policy: go at A, wait at B (1 each). Checks: $\frac{11}{4} \geq \frac{5}{2}$ and $\frac{7}{2} \geq 3$ (1 each). These are action values under the original π after the first action, not the values of the new policy; do not label them v^*.

(e) (6 points) Let π' be the greedy policy from part (d). Part (d) checks that using π' for one step and then following π is at least as good as following π throughout: $v^\pi(s) \le q^\pi(s, \pi'(s))$ at both states. Explain why using π' for two steps and then following π is at least as good as using it for one step. (Hint: apply the same inequality at the state reached after the first step.) What happens as the number of steps using π' grows, and what does this imply about $v^{\pi'}$ and v^π? Then state the main difference between a policy-iteration cycle and a value-iteration sweep.

Solution: $q^\pi(s, \pi'(s))$ is the expected value of $R_{t+1} + \gamma v^\pi(S_{t+1})$ when $\pi'(s)$ is taken. Replacing $v^\pi(S_{t+1})$ by $q^\pi(S_{t+1}, \pi'(S_{t+1}))$ lowers no term (at a terminal successor both continuations are zero), and probabilities and γ are nonnegative, so the average does not fall; the result is two steps of π' and then π. Repeating gives k steps of π' followed by π, a sequence that never decreases. Values are bounded and the remaining part is discounted by $\gamma^k \to 0$, so the limit is $v^{\pi'}$ and $v^{\pi'}(s) \ge v^\pi(s)$. Here A: $\frac{5}{2}, \frac{11}{4}, \frac{23}{8}, \ldots \to 3$ and B: $3, \frac{7}{2}, \frac{15}{4}, \ldots \to 4$ (not required). Policy iteration evaluates a fixed policy (exactly or nearly) and then improves it greedily; value iteration applies the maximizing backup in every sweep without first completing a fixed-policy evaluation. Rubric: two-step argument 2 (inequality at each successor 1, averaging preserves it 1), growing k 1, vanishing tail and conclusion 1, PI 1, VI 1. Accept a verbal argument. Award the reasoning points independently of a mistaken action choice in (d) when the student correctly uses the supplied inequality.

4. (20 points) A fixed policy π produced these two complete episodes. Each arrow is labelled by the reward received on that transition. The model is unknown. Discount $\gamma = \frac{1}{2}$.

   - Episode 1: $A \xrightarrow{0} A \xrightarrow{2} B \xrightarrow{4}$ terminal
   - Episode 2: $A \xrightarrow{0} B \xrightarrow{2}$ terminal

   The actions are omitted because this question only evaluates π.

   (a) (7 points) Use **first-visit** Monte Carlo sample averages over both episodes to estimate $V(A)$ and $V(B)$. Then give the **every-visit** Monte Carlo estimate of $V(A)$. For each estimate, list the returns you included.

   Solution: Episode 1 returns: $G_2 = 4$, $G_1 = 2 + \frac{1}{2} \cdot 4 = 4$, $G_0 = 0 + \frac{1}{2} \cdot 4 = 2$. Episode 2: $G_1 = 2$, $G_0 = 1$. First-visit: $V(A) = (2 + 1)/2 = \frac{3}{2}$, $V(B) = (4 + 2)/2 = 3$. Every-visit: $V(A) = (2 + 4 + 1)/3 = \frac{7}{3}$ (episode 1's second visit to A has return 4). Rubric: first-visit included returns 2, first-visit estimates 2, every-visit included returns for A 1, its estimate 1, correct visit counts 1.

   (b) (7 points) Independently of part (a), initialize $V(A) = 2$ and $V(B) = 0$. With $\alpha = \frac{1}{2}$, apply TD(0) updates online, in time order, through **episode 1 only**. For each transition, give the target and the updated value.

Transition	Target U_t	Updated entry	$(V(A), V(B))$ after
$A \xrightarrow{0} A$	$0 + \frac{1}{2} \cdot 2 = 1$	$V(A) = 2 + \frac{1}{2}(1 - 2) = \frac{3}{2}$	$(\frac{3}{2}, 0)$
$A \xrightarrow{2} B$	$2 + \frac{1}{2} \cdot 0 = 2$	$V(A) = \frac{3}{2} + \frac{1}{2}(2 - \frac{3}{2}) = \frac{7}{4}$	$(\frac{7}{4}, 0)$
$B \xrightarrow{4}$ terminal	4	$V(B) = 0 + \frac{1}{2}(4 - 0) = 2$	$(\frac{7}{4}, 2)$

   Solution: The first target and the subtracted estimate both use the pre-update value $V(A) = 2$ (self-transition). Rubric: each target 1, each update from the student's current table 1, chronological use of the reset table 1.

(c) (6 points) This part uses **new data** and $\gamma = 1$. The data are four complete episodes:

- twice: $C \xrightarrow{+1} D \xrightarrow{0}$ terminal;
- twice: $D \xrightarrow{+2}$ terminal.

Using the batch convention from class (repeated passes over this fixed data, increments summed over a pass with the table held fixed, and a small enough step size to converge), what do batch Monte Carlo and batch TD(0) converge to for $V(C)$ and $V(D)$? Which answer would you trust if you knew the process was Markov, and why might you doubt that here?

Solution: Both methods: $V(D) = (0 + 0 + 2 + 2)/4 = 1$ (1). Batch MC: $V(C) = 1$, the mean of its observed returns (1). Batch TD: $V(C) = 1 + V(D) = 2$ (2). If the process is Markov, TD's answer uses all four outcomes observed from D (certainty equivalence) (1). Here D gave 0 every time it followed C and 2 every time it started an episode, which hints that "D" may hide relevant history; then pooling its outcomes can mislead. Two episodes of each kind are little evidence either way (1). For the "trust" point, also credit a reasoned answer that neither estimate is guaranteed to be more accurate from four episodes. Do not require a claim that the data prove the process is not Markov.

5. (14 points) This is a new independent episode, with $\gamma = \frac{1}{2}$ and time indices $0, 1, 2, 3$:

$$A \xrightarrow{2} B \xrightarrow{0} C \xrightarrow{4} \text{terminal}.$$

Hold the value table fixed throughout: $V(A) = 0$, $V(B) = 4$, $V(C) = 6$, and terminal value 0. Targets are for the visit to A at time 0; do not update the table between calculations.

(a) (4 points) Expand the return recursion twice and replace the unobserved return after the second transition with its value estimate. Write the resulting two-step target $G_{0:2}$ symbolically before inserting numbers.

Solution: $G_0 = R_1 + \gamma G_1 = R_1 + \gamma R_2 + \gamma^2 G_2$; replacing G_2 with $V(S_2)$ gives $G_{0:2} = R_1 + \gamma R_2 + \gamma^2 V(S_2)$. Rubric: first expansion 1, second expansion with correct discount powers 1, replacement at the second successor 1, correct target 1.

(b) (6 points) Calculate the one-step target $G_{0:1}$, the two-step target $G_{0:2}$ and the full return G_0. For the full return, indicate why there is no additional bootstrap term.

Solution: $G_{0:1} = 2 + \frac{1}{2} \cdot 4 = 4$. $G_{0:2} = 2 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 6 = \frac{7}{2}$. $G_0 = 2 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 4 = 3$: the episode ended, so there is no continuation to bootstrap from. Rubric: each target, expression 1 and result 1.

(c) (1 point) If this episode arrives one transition at a time, at which time index does the two-step target for A first become available? Must the learner wait until termination here?

Solution: At time 2, after observing $R_2 = 0$ and $S_2 = C$; no need to wait for termination at time 3.

(d) (3 points) A student sets $\lambda = 1$ in the finite episodic forward λ-return to remove bootstrapping, then concludes that the target must be more accurate than TD(0). Assess both claims briefly.

Solution: At $\lambda = 1$ the λ-return is the complete return and does not bootstrap (1). The accuracy claim is false (1): removing the table from the target does not remove sampling variability, so the error need not be lower and a good bootstrapped estimate can help (1).

Several Equations (semantics and context are not given on purpose)

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}, \qquad G_t = R_{t+1} + \gamma G_{t+1}, \qquad G_T = 0$$

$$v^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad q^\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

$$v^\pi(s) = \sum_a \pi(a \mid s)\, q^\pi(s,a)$$

$$q^\pi(s,a) = \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, v^\pi(s')\big]$$

$$v^*(s) = \max_a \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, v^*(s')\big], \qquad \pi^*(s) \in \arg\max_a q^*(s,a)$$

$$V_{k+1}(s) = \sum_a \pi(a \mid s) \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, V_k(s')\big]$$

$$V_{k+1}(s) = \max_a \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, V_k(s')\big]$$

$$V(S_t) \leftarrow V(S_t) + \alpha\big[U_t - V(S_t)\big]$$

$$U_t = G_t, \qquad U_t = R_{t+1} + \gamma V(S_{t+1})$$

$$V(s) \leftarrow V(s) + \frac{1}{N(s)}\big[G_t - V(s)\big], \qquad S_t = s$$

$$G_t^\lambda = (1-\lambda) \sum_{n=1}^{T-t-1} \lambda^{n-1} G_{t:t+n} + \lambda^{T-t-1} G_t, \qquad 0 \le \lambda \le 1, \quad t < T$$